

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
Due Date	30/3/26		

Experiment 8: Effect of PCA on Regression and Classification Models

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1 Objective

To study the impact of PCA on regression and classification models.

2 Dataset

- Samples: 1000
- Features: Academic attributes
- Targets:
 - Regression: Final Score
 - Classification: Performance Level

3 EDA

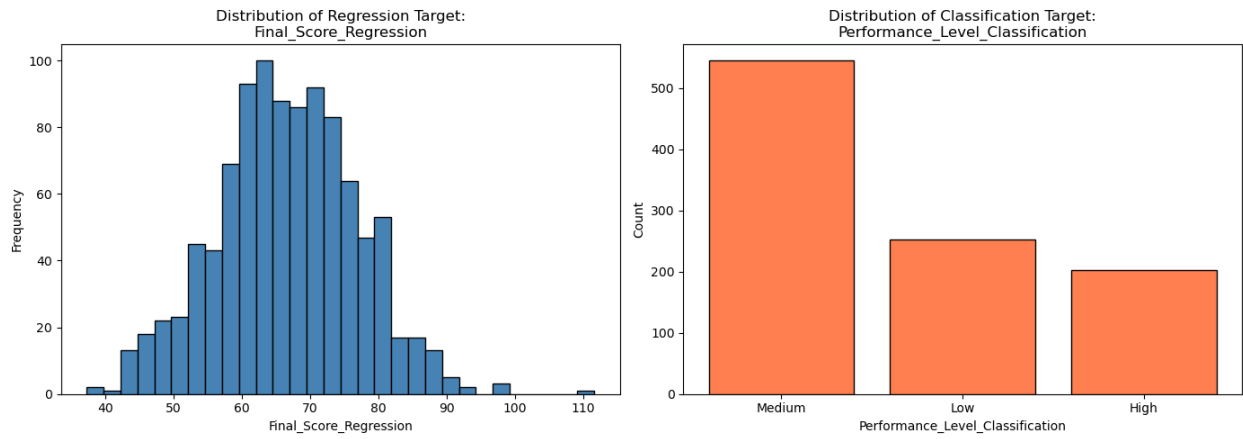


Figure 1: Target Distribution

4 PCA Analysis

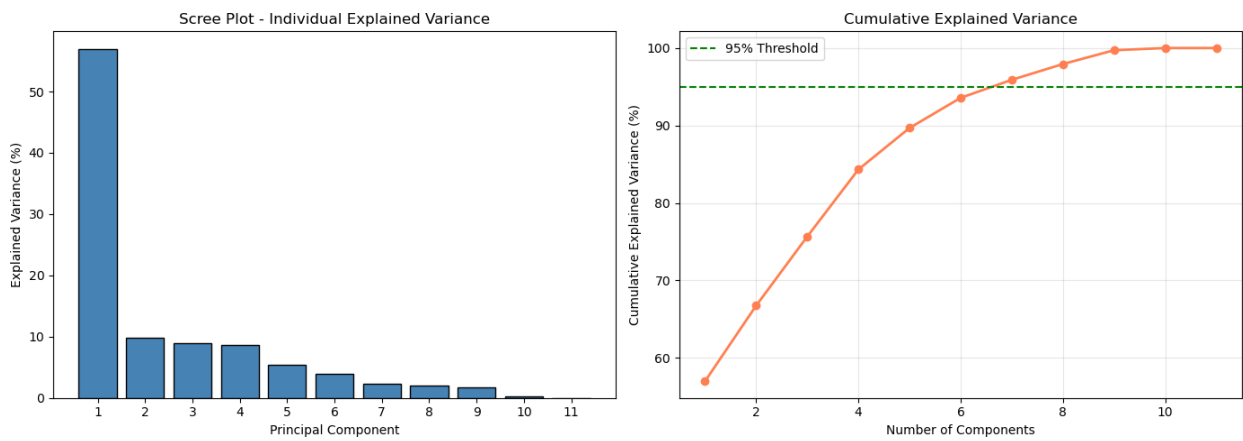


Figure 2: Scree Plot and Cumulative Variance

5 PCA Summary

Table 1: PCA Summary

Components Chosen	Explained Variance (%)	Justification
7	95.90	Retains 95% variance, reduces dimensionality and mult

6 Regression Results

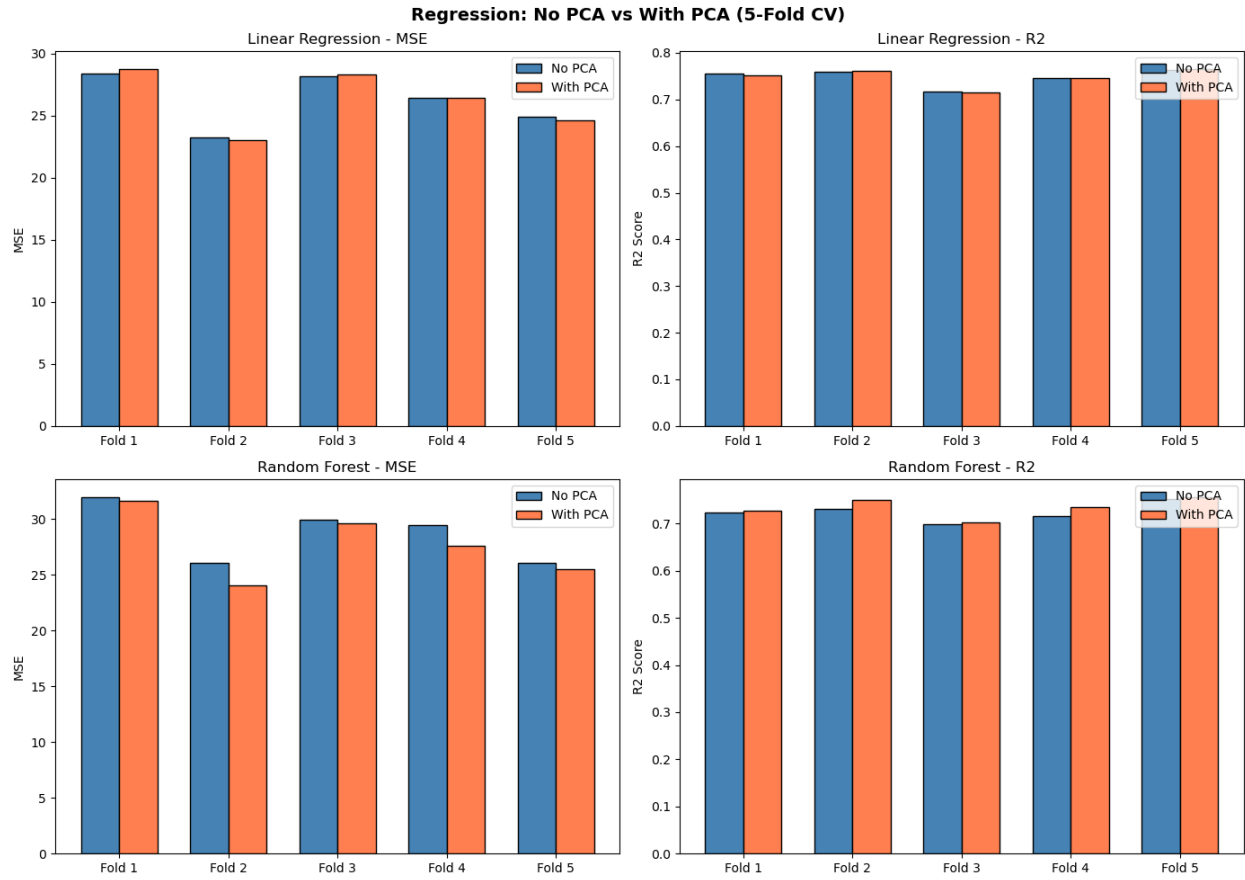


Figure 3: Regression Performance (No PCA vs PCA)

Table 2: 5-Fold CV Results – Regression

Model	Metric	No PCA	With PCA	Observation
Linear Regression	MSE	26.2338 ± 1.9532	26.2251 ± 2.1705	PCA slightly reduces error
Linear Regression	R2	0.7479 ± 0.0165	0.7481 ± 0.0176	Slight improvement
Random Forest	MSE	28.7219 ± 2.3318	27.7007 ± 2.7462	Reduced error
Random Forest	R2	0.7241 ± 0.0172	0.7342 ± 0.0194	Improved fit

7 Classification Results

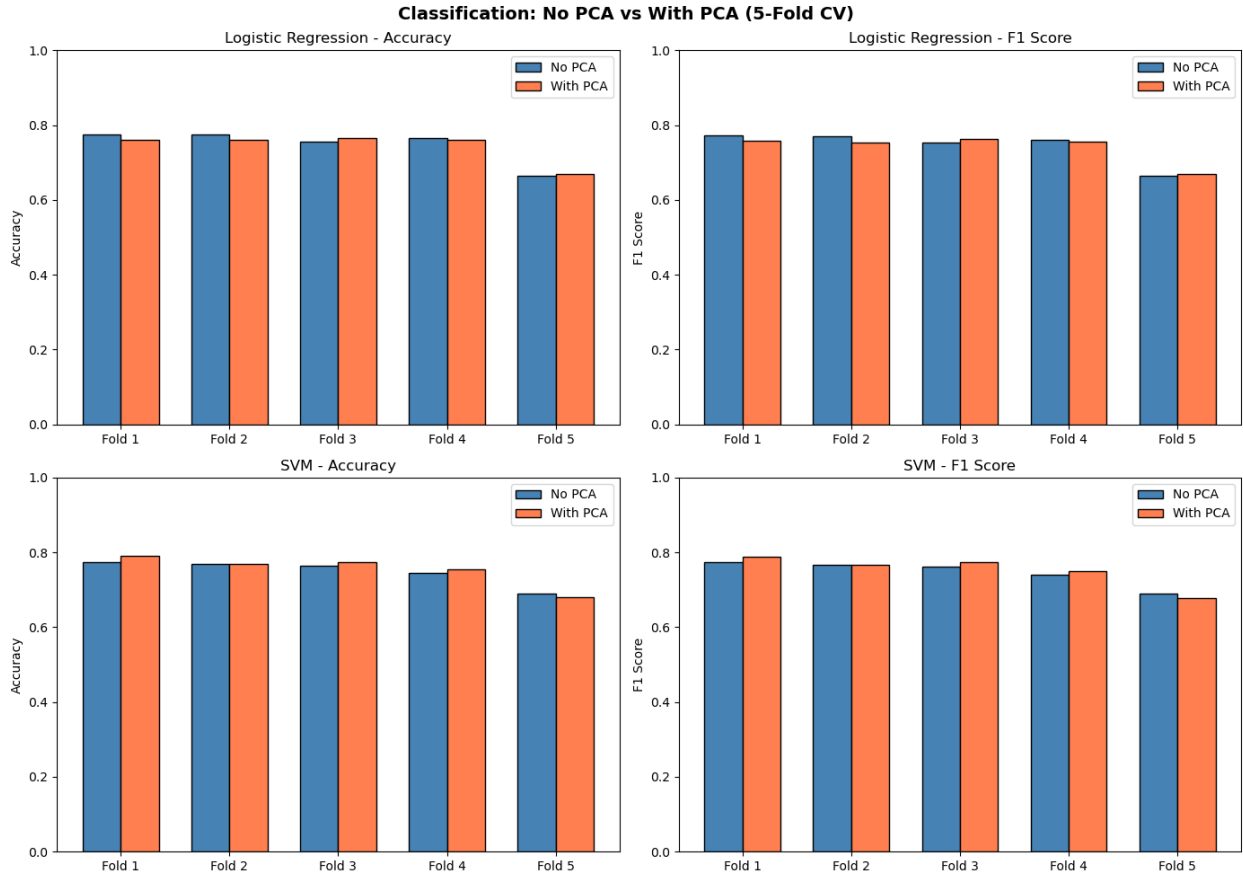


Figure 4: Classification Performance (No PCA vs PCA)

Table 3: 5-Fold CV Results – Classification

Model	Metric	No PCA	With PCA	Observation
Logistic Regression	Accuracy	0.7470 $\pm$ 0.0417	0.7430 $\pm$ 0.0366	Minimal change
Logistic Regression	F1	0.7441 $\pm$ 0.0409	0.7401 $\pm$ 0.0358	Minimal change
SVM	Accuracy	0.7490 $\pm$ 0.0312	0.7540 $\pm$ 0.0387	Slight improvement
SVM	F1	0.7466 $\pm$ 0.0308	0.7513 $\pm$ 0.0383	Slight improvement

8 Observations

- PCA reduces dimensionality and multicollinearity.
- Linear models benefit slightly from PCA.
- Random Forest shows moderate improvement.
- Classification models show minimal change.

9 Conclusion

PCA improves model performance by reducing noise and multicollinearity. It benefits linear models more than tree-based models. However, excessive reduction may lead to information loss.